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Paper Code : PEC-IT601A Advanced Algorithms

UPID : 006589

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) An independent subset in a matroid M is said to be_____if it is not contained in any larger independent subset of M .

(II) Strassen's algorithm uses _____strategy to multiply two matrices.

(III) Floyd Warshall algorithm uses dynamic programming . True or False ?

(IV) A problem is _____if there is an algorithm that says yes if the answer is yes, and no otherwise.

(V) All the maximal independent subsets in a matroid have the_____ size .

(VI) _____are the problems that can be solved in exponential time complexity.

(VII) What theorem states that A matching M in G has maximum size iff G has no augmented path ?

(VIII) Which matrix is the inverse of an upper triangular?

(IX) The_____ problem asks to find the shortest path in a graph with the condition of visiting all the nodes only one time and returning to the origin city.

(X) An algorithm that uses random numbers to decide what to do next anywhere in its logic is called a _____.

(XI) _____for Directed Acyclic Graph (DAG) is a linear ordering of vertices such that for every directed edge <u,v> vertex u comes before v in the ordering.

(XII) _____are all edges that can go any vertices in the same depth first tree as long as one is the ancestor of another or between vertices in different depth first tree.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Write an algorithm for topological sort using DFS. Use the same to solve the graph in fig.2 . [5]

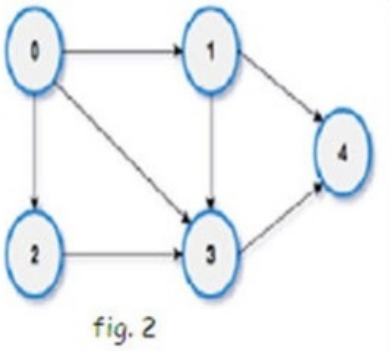


fig. 2

3. Find the shortest path and their cost between every pair of vertices for the problem instance [5]

$$\begin{bmatrix} 0 & 4 & 11 \\ 6 & 0 & 2 \\ 3 & inf & 0 \end{bmatrix}$$

4. Prove that a forest F(VF, EF) has |VF|-|EF| trees. [5]

5. Given the instance for Chinese Remainder theorem where n1=5, n2=7, n3=8 and a=89, and a (a1,a2,a3). Find n,a1,a2,a3. [5]

6. Derive the recurrence relation of Strassen's matrix multiplication algorithm using divide and conquer and analyse its complexity. [5]

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Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Give the basic outline of a genetic algorithm. [5]

   (b) Suggest some suitable searching and sor ting algorithms and briefly explain their contribu tions in emerging trends of computing. [10]

8. Write short note on I) Schonhage-Strassen Integer Mul tiplication algorithm ii) Modulo representa tion of numbers iii) Solving TSP with dynamic programming. [15]

9. Define optimization and decision problem with examples. Define P, NP, NP-hard and Np-complete class of problems. How are these classes related to each other? State Cook's theorem. Differen tiate between deterministic and nondeterministic algorithms. [15]

10. What are triangular matrices? What is the idea behind LUP decomposi tion? State its advantages. Show that the LUP decomposition give same solution as the original matrix equation. [15]

11. State clique decision problem. Show that the clique decision problem is NP-complete. [15]

*** END OF PAPER ***

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